Initially, amplifiers were made of vacuum tubes and currently almost all are solid state. Although, there are some who still prefer the original tube sound. Amplifiers mostly support multiple input types as audio sources like CD players, digital audio players, cassette players etc.


A typical amplifier in a hi-fi music system will let you play with a lot of controls

To better understand what exactly it does, consider the fact that the input signal to the amplifier cannot be more than a few hundred microwatts. The output, on the other hand, could vary between a few watts for household electronic equipment to tens of thousands of watts for a rock concert. Power amplifiers are available as standalone units for audiophiles and high end listening needs. On the other hand, most consumer devices ship with built in amplifiers and even DACs.

DAC-ADC

Digital to Analog and Analog to Digital converters have played one of the most crucial roles in pushing audio, as well as other forms of digital media, out to the masses. Consider a typical phone call taking place over a long distance. The microphone at the caller's end converts his or her voice into an analog electrical signal, which then is converted into a digital stream by the ADC. This digital stream treats the audio data as just another piece of digital data. On the receiving end, the audio is reassembled in the form of a digital audio stream. Now, a DAC converts this back to an analog electrical signal, which is fed to an amplifier that drives a loudspeaker to produce sound.

DACs are obviously found in music players, sound processing units and more. Specialist stand alone audio DACs are also found with high quality audio systems.

Microphone

A microphone is a device that is used to capture sound, rather than play it back. Most microphones today use electromagnetic induction (dynamic microphones), capacitance change (condenser microphones) or piezoelectricity (piezoelectric microphones) to produce an electrical signal from